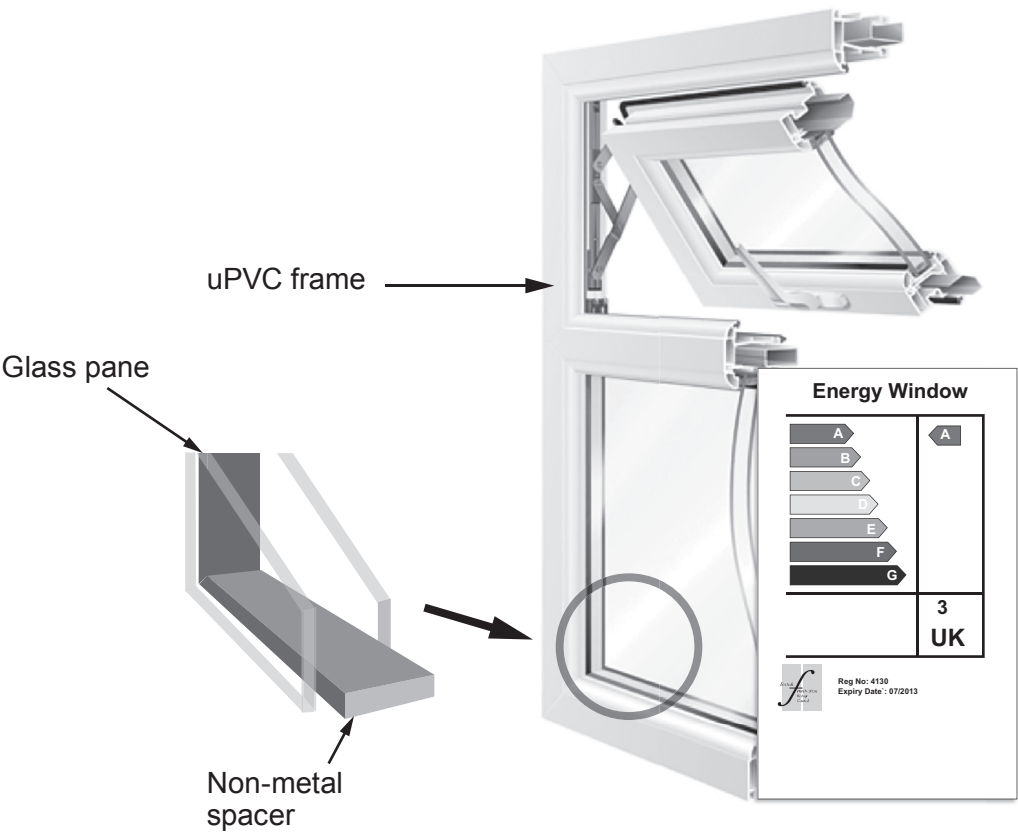


4. All properties lose heat through their windows and roof. Energy-efficient installations such as uPVC windows and loft insulation keep homes warmer as well as reducing energy bills.
- (a) A homeowner replaces aluminium framed double glazed windows which contain metal spacers between the panes of glass with uPVC windows. This new frame is made of uPVC and the window panes are separated by non-metal spacers.



- (i) Explain why the aluminium window loses heat at a faster rate than the new uPVC window. Include reference to particles in your answer. [4]

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(ii) The photograph shows an uninsulated loft.



Explain how heat is lost through the roof of a house when the loft is not insulated.
You should refer to the ceiling, the loft space and the roof tiles in your answer. [3]

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(b) Explain why reducing energy losses from our homes benefits the environment. [3]

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